Answer Key - Biology - Graduate Level Assessment 3 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Water temperature alone without hormonal influence; B) The age of the crustacean; C) Environmental factors; D) Social interactions with other crustaceans; E) Diet of the crustacean
   Note: The process of molting in crustaceans is primarily triggered by water temperature changes, as it influences metabolic rates and physiological conditions necessary for shedding their exoskeleton, independent of hormonal factors.

2. **Answer:** A
   Options: A) $2.5 \times 10^{-}5; B)1.5 \times 10^{-}5; C)3.0 \times 10^{-}5; D)4.0 \times 10^{-}5; E)5.0 \times 10^{-}5$
   $Note:Thecorrectanswerisderivedfromtherelationshipbetweenthefrequencyofthedominanttrait(1/8$
   (1 - w) / w, where q is the frequency of the recessive allele and w is the fitness of the dominant allele.

3. **Answer:** A
   Options: A) Man evolved directly from monkeys; B) Man evolved from a branch of primates that includes the modern lemur; C) Man evolved from Parapithecus; D) Man evolved from a direct line of increasingly intelligent monkeys; E) Man evolved from Neanderthals who were a separate species from monkeys
   Note: The statement is incorrect because humans and modern monkeys share a common ancestor rather than evolving directly from one another, highlighting the concept of divergent evolution in the study of primate phylogeny.

4. **Answer:** C
   Options: A) tendons; B) skin; C) muscle; D) blood; E) cartilage
   Note: Muscle is classified as a type of tissue distinct from connective tissue, as it primarily functions in movement through contraction, whereas connective tissues serve to support, bind, and protect other tissues and organs in the body.

5. **Answer:** A
   Options: A) Complete absence of segmentation; B) Hindrance in the development of the nervous system; C) Inability to undergo metamorphosis; D) Absence of a group of contiguous segments; E) Tumor formation in imaginal discs
   Note: The mutation of homeotic cluster genes disrupts the normal patterning and segmentation of Drosophila, often leading to the complete absence of segmentation by failing to specify distinct body segments during development.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: The general practice selection process is predictive of future performance in the MRCGP.

7. **Answer:** False
   Options: A) True; B) False
   Note: Improving hospital efficiency by shortening length of stay does not appear to result in increased rates of readmission or numbers of physician visits within 30 days after discharge from hospital. Research is needed to identify optimal lengths of stay and expected readmission rates.

8. **Answer:** True
   Options: A) True; B) False
   Note: Font influenced pregnant women's ratings of intervention complexity.

9. **Answer:** False
   Options: A) True; B) False
   Note: The incidence of abdominal injury in intoxicated, hemodynamically stable, blunt trauma patients with a normal abdominal examination and normal mentation is low. Physical examination and attention to clinical risk factors allow accurate abdominal evaluation without CT.

10. **Answer:** False
    Options: A) True; B) False
    Note: CLM resection following second-line PCT, after oncosurgically favourable selection, could bring similar OS compared to what observed after first-line. For initially unresectable patients, OS or DFS is comparable between first- and second-line PCT. Surgery should not be denied after the failure of first-line chemotherapy.

Long Form Response

11. **Answer:** The courtship behavior of the common tern is characterized by the male's presentation of fish to potential mates, which serves as a display of his foraging ability and overall fitness. This ritual, while seemingly aimed at securing a mate, also functions as a form of entertainment for both the male and female terns, underscoring the social dynamics at play in avian courtship. Such behaviors can enhance reproductive success by allowing females to assess the quality of males, thereby ensuring that they select partners capable of providing resources for offspring. Moreover, the broader significance of these courtship rituals extends beyond individual pair bonding; they play a crucial role in the evolutionary strategies of avian species, influencing mate selection and ultimately contributing to the genetic diversity and adaptability of populations. In this context, the courtship of common terns exemplifies how seemingly simple behaviors can have profound implications for the survival and evolution of species.

Note: correct because it effectively highlights the significance of the male common tern's fish presentation as a display of fitness that plays a crucial role in female mate selection, while also emphasizing the social dynamics and entertainment value of courtship rituals in enhancing reproductive success among birds.

12. **Answer:** The biological clock of fiddler crabs can be reset through specific environmental manipulations, such as artificially creating cycles of light and darkness or exposing the crabs to extreme temperatures, like ice water. These interventions exploit the inherent plasticity of the crabs' circadian rhythms, which are sensitive to external cues. Light exposure, particularly, can influence melatonin production and other hormonal pathways that regulate color change, while cold exposure may trigger physiological stress responses that affect metabolic processes linked to pigmentation. By altering these environmental factors, researchers can induce a reversal in the crabs' color change patterns, illustrating the interplay between environmental stimuli and physiological adaptations in the regulation of circadian rhythms.
    Note: correct because it identifies the role of environmental factors such as light and temperature in resetting the fiddler crabs' biological clock, which in turn affects hormonal regulation and the color change mechanism, illustrating the influence of external cues on circadian rhythms.

End of Answer Key